EI331 Signals and Systems

Lecture 19

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1. Convolution Property of DTFT

2. Multiplication Property of DTFT

3. Systems Described by Linear Constant-coefficient Difference Equations

4. Sampling of DT Signals

5. DT Processing of CT Signals

Convolution Property of DTFT

$$(x * h)[n] \xleftrightarrow{\mathcal{F}} X(e^{j\omega})H(e^{j\omega})$$

NB. Dual of multiplication property of CTFS

$$x(t)y(t) \xleftrightarrow{\mathcal{FS}} (\hat{x} * \hat{y})[k]$$

NB. Similar to CTFT, applicable when formula is well-defined

Proof.

$$\begin{aligned}
\mathcal{F}\{x * h\}(e^{j\omega}) &= \sum_{n\in\mathbb{Z}} (x * h)[n]e^{-j\omega n} = \sum_{n\in\mathbb{Z}} \sum_{m\in\mathbb{Z}} x[m]h[n-m]e^{-j\omega n} \\
&= \sum_{m\in\mathbb{Z}} x[m] \sum_{n\in\mathbb{Z}} h[n-m]e^{-j\omega n} \\
&= \sum_{m\in\mathbb{Z}} x[m]e^{-j\omega m}H(e^{j\omega}) \\
&= X(e^{j\omega})H(e^{j\omega})
\end{aligned}$$

Frequency Response of LTI System

LTI system T

- fully characterized by impulse response h

$$y = T(x) = h * x$$

- also fully characterized by frequency response $H = \mathcal{F}\{h\}$, if H is well-defined
  - BIBO stable system, $h \in \ell_1$
  - other systems: accumulator $h = u$, ...

Typically, convolution property implies

$$Y = \mathcal{F}\{y\} = HX = \mathcal{F}\{h\}\mathcal{F}\{x\}$$

Instead of computing $x * h$, can do

$$y = \mathcal{F}^{-1}(\mathcal{F}\{h\}\mathcal{F}\{x\})$$

Example

Response of LTI system with impulse response $h[n] = a^n u[n]$ to input $x[n] = b^n u[n]$, where $|a| < 1, |b| < 1$

Method 1. Direct convolution $y = x * h$

Method 2. Solve difference equation with initial rest condition

$$y[n] - ay[n-1] = b^n u[n]$$

Method 3. Fourier transform.

$$H = \frac{1}{1 - ae^{-j\omega}},\ X = \frac{1}{1 - be^{-j\omega}} \implies Y = \frac{1}{(1 - ae^{-j\omega})(1 - be^{-j\omega})}$$

If $a \neq b$, $Y(e^{j\omega}) = \frac{a/(a-b)}{1-ae^{-j\omega}} - \frac{b/(a-b)}{1-be^{-j\omega}}$, $\quad y[n] = \frac{1}{a-b}(a^{n+1} - b^{n+1})u[n]$

If $a = b$, $Y(e^{j\omega}) = \frac{d}{da}\left(\frac{a}{1-ae^{-j\omega}}\right), y[n] = \frac{d}{da}a^{n+1}u[n] = (n+1)a^n u[n]$

Or use $Y(e^{j\omega}) = \frac{j}{a}e^{j\omega}\frac{d}{d\omega}\left(\frac{a}{1-ae^{-j\omega}}\right)$ and differentiation property

Example

Response of LTI system with impulse response $h[n] = a^n u[n]$ to input $x[n] = \cos(\omega_0 n)$, $|a| < 1$.

Frequency response

$$H(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$$

Method 1. Use eigenfunction property

$$x[n] = \frac{1}{2}e^{j\omega_0 n} + \frac{1}{2}e^{-j\omega_0 n}$$

$$\begin{aligned} y[n] &= \frac{1}{2}H(e^{j\omega_0})e^{j\omega_0 n} + \frac{1}{2}H(e^{-j\omega_0})e^{-j\omega_0 n} = \text{Re}\,\frac{e^{j\omega_0 n}}{1 - ae^{-j\omega_0}} \\ &= \frac{1}{\sqrt{1 - 2a\cos\omega_0 + a^2}}\cos\left(\omega_0 n - \arctan\frac{a\sin\omega_0}{1 - a\cos\omega_0}\right) \end{aligned}$$

Example

Method 2. Use Fourier transform of X

$$X(e^{j\omega}) = \sum_{k=-\infty}^{\infty} [\pi\delta(\omega - \omega_0 + 2k\pi) + \pi\delta(\omega + \omega_0 + 2k\pi)]$$

$$Y(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \pi H(e^{j(\omega_0 - 2k\pi)})\delta(\omega - \omega_0 + 2k\pi)$$

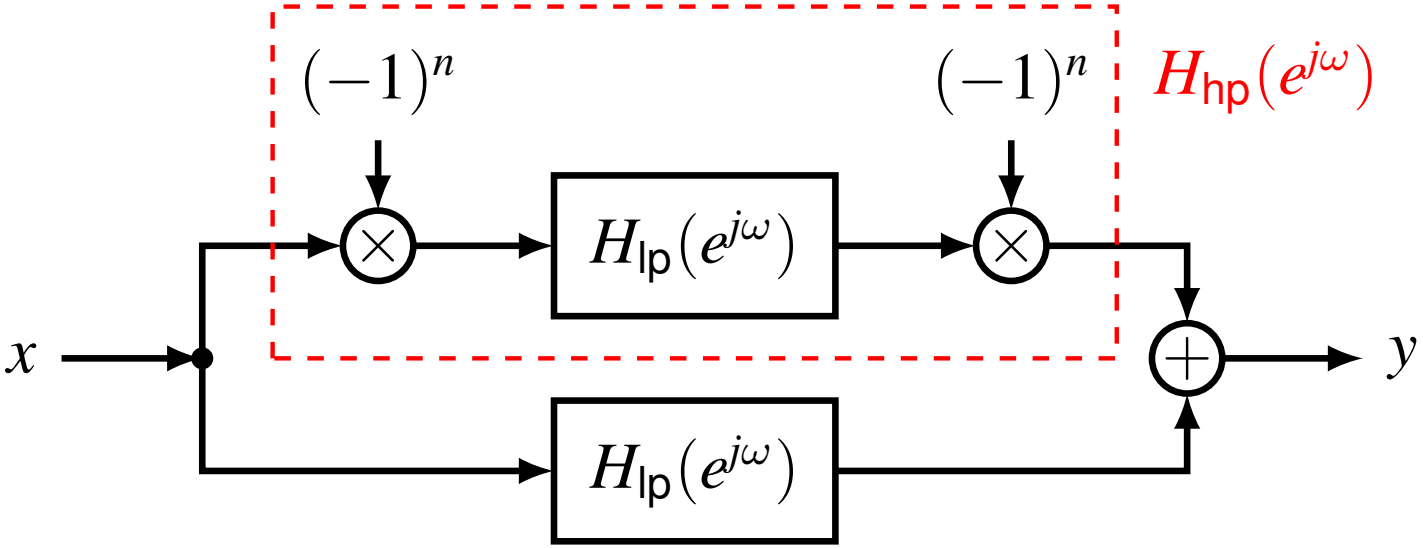
$$+ \sum_{k=-\infty}^{\infty} \pi H(e^{-j(\omega_0 + 2k\pi)})\delta(\omega + \omega_0 + 2k\pi)$$

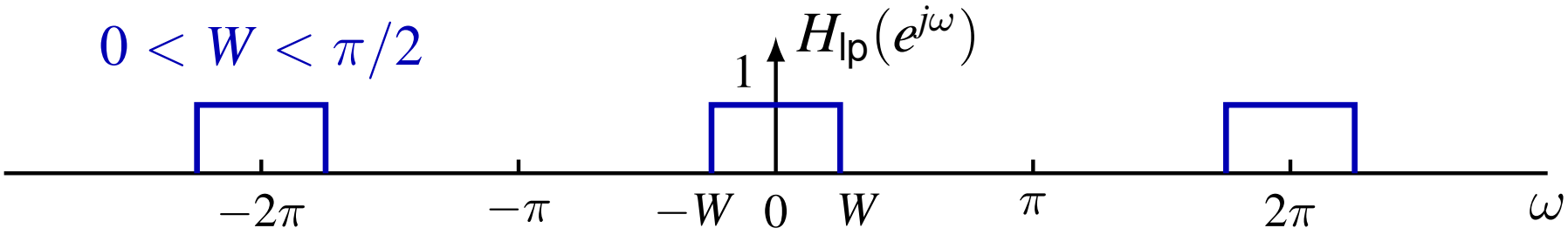
$$= H(e^{j\omega_0}) \sum_{k=-\infty}^{\infty} \pi\delta(\omega - \omega_0 + 2k\pi)$$

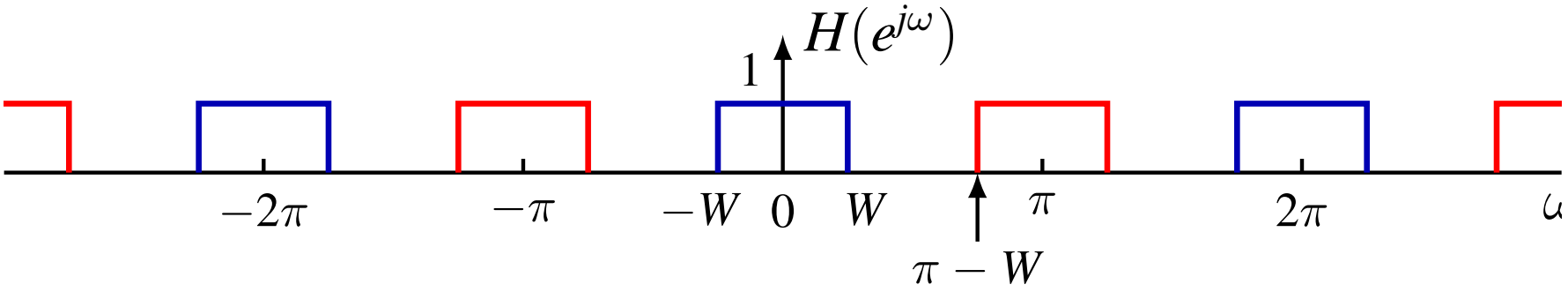
$$+ H(e^{-j\omega_0}) \sum_{k=-\infty}^{\infty} \pi\delta(\omega + \omega_0 + 2k\pi)$$

$$y[n] = \frac{1}{2}H(e^{j\omega_0})e^{j\omega_0 n} + \frac{1}{2}H(e^{-j\omega_0})e^{-j\omega_0 n}$$

Example: Ideal Bandstop Filter







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Multiplication Property of DTFT

$$x[n]y[n] \xleftrightarrow{\mathcal{F}} \frac{1}{2\pi} X \circledast Y = \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta$$

NB. Dual of periodic convolution property of CTFS

$$x \circledast y \xleftrightarrow{\mathcal{FS}} T\hat{x}\hat{y}$$

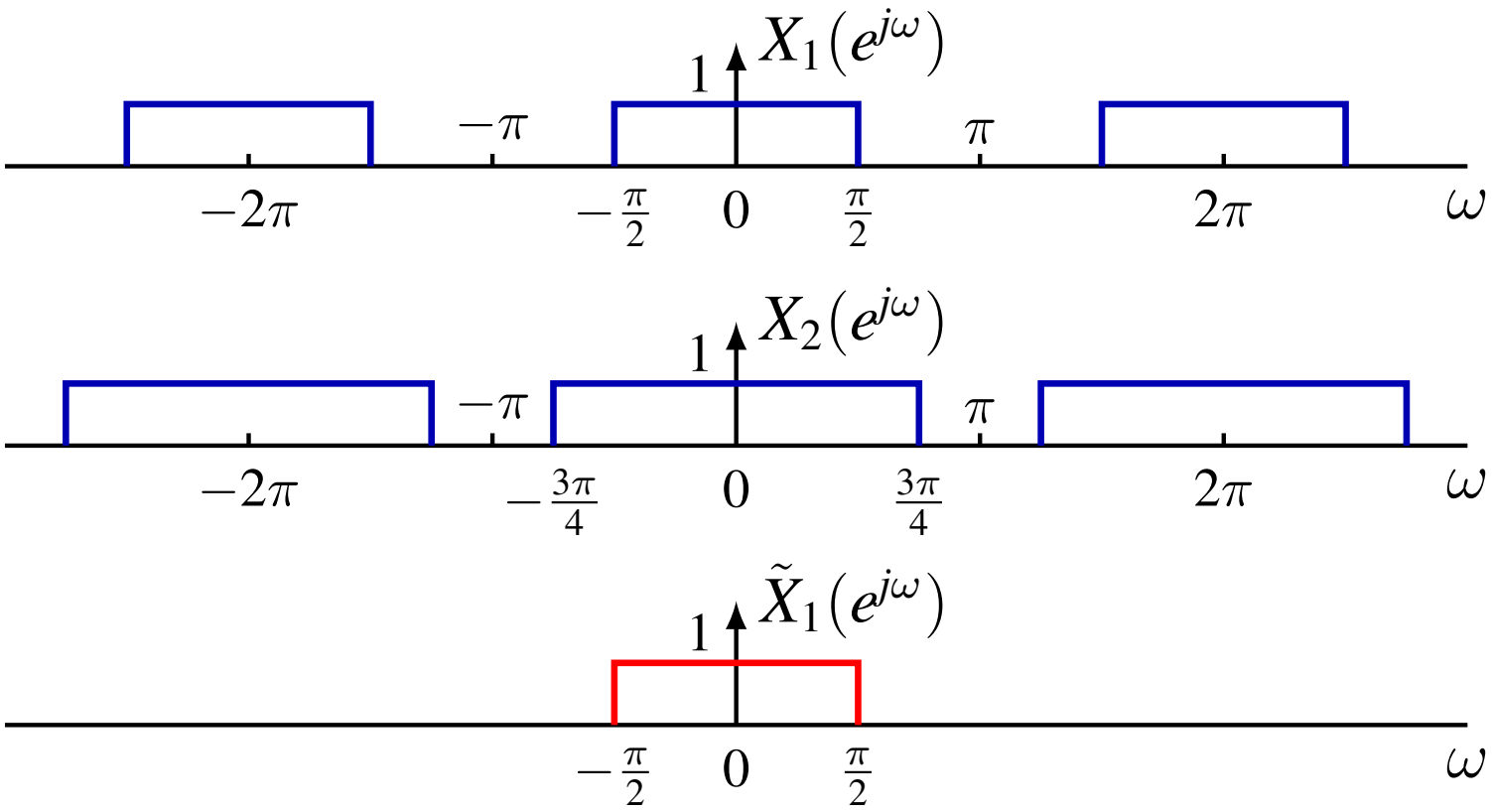
Proof.

$$\begin{aligned}
\mathcal{F}\{xy\}(e^{j\omega}) &= \sum_{n\in\mathbb{Z}} x[n]y[n]e^{-j\omega n} = \sum_{n\in\mathbb{Z}} \left(\frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) e^{j\theta n} d\theta \right) y[n]e^{-j\omega n} \\
&= \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) \left(\sum_{n\in\mathbb{Z}} y[n]e^{-j(\omega-\theta)n} \right) d\theta \\
&= \frac{1}{2\pi} \int_{2\pi} X(e^{j\theta}) Y(e^{j(\omega-\theta)}) d\theta
\end{aligned}$$

Example

$$x[n] = x_1[n]x_2[n], \text{ where } x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}, \; x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$$

$$X = \tfrac{1}{2\pi} X_1 \underset{\text{periodic}}{\circledast} X_2 = \tfrac{1}{2\pi} \tilde{X}_1 \underset{\text{aperiodic}}{*} X_2$$

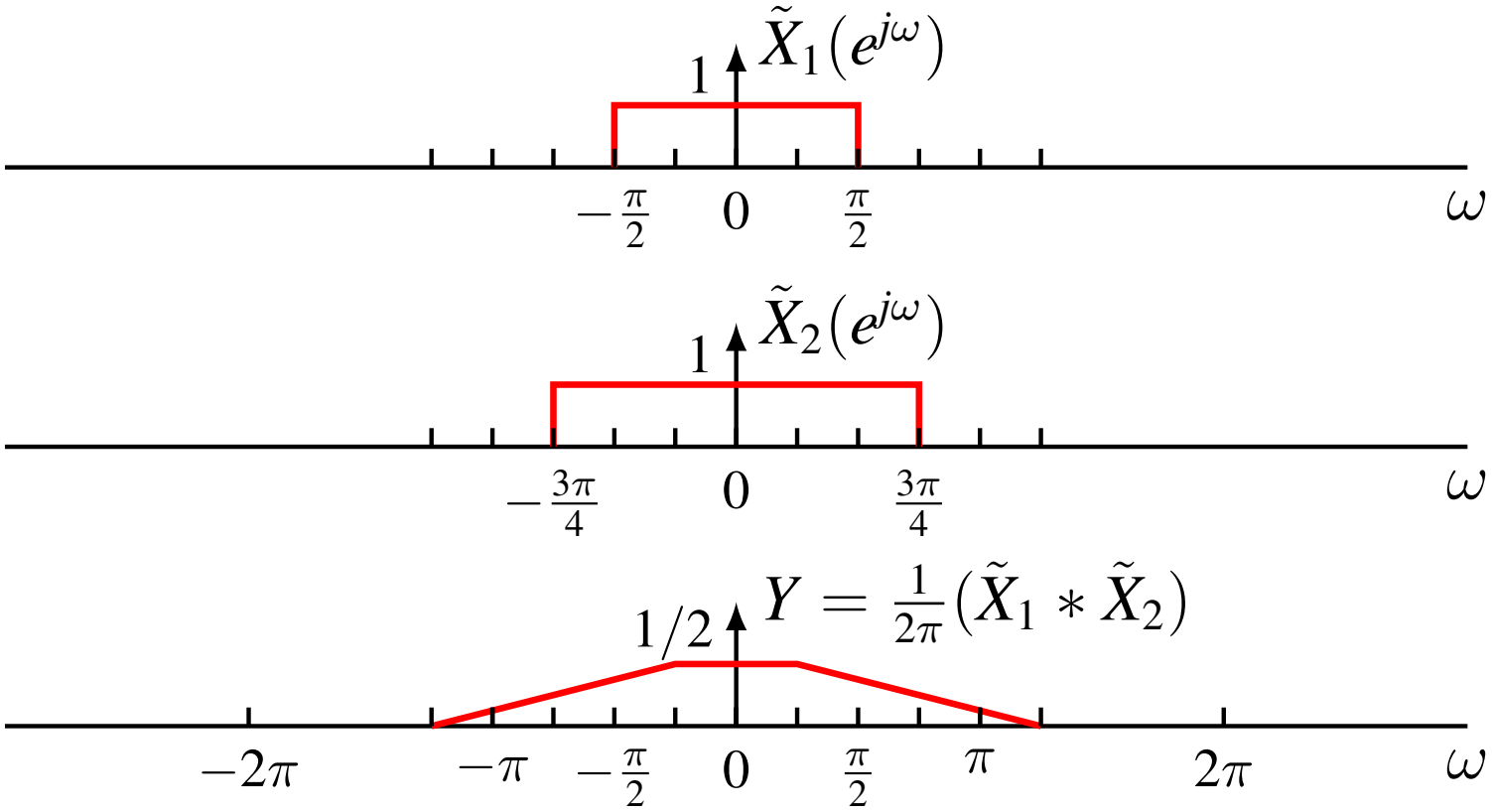


Example

$$x[n] = x_1[n]x_2[n], \text{ where } x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}, \; x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$$

$$X = \tfrac{1}{2\pi} X_1 \circledast X_2 = \tfrac{1}{2\pi} \tilde{X}_1 * X_2 = \tfrac{1}{2\pi} \sum_{k=-\infty}^{\infty} \tau_{2k\pi}(\tilde{X}_1 * \tilde{X}_2)$$

periodic aperiodic aperiodic



Example

$$x[n] = x_1[n]x_2[n], \text{ where } x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}, \; x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$$

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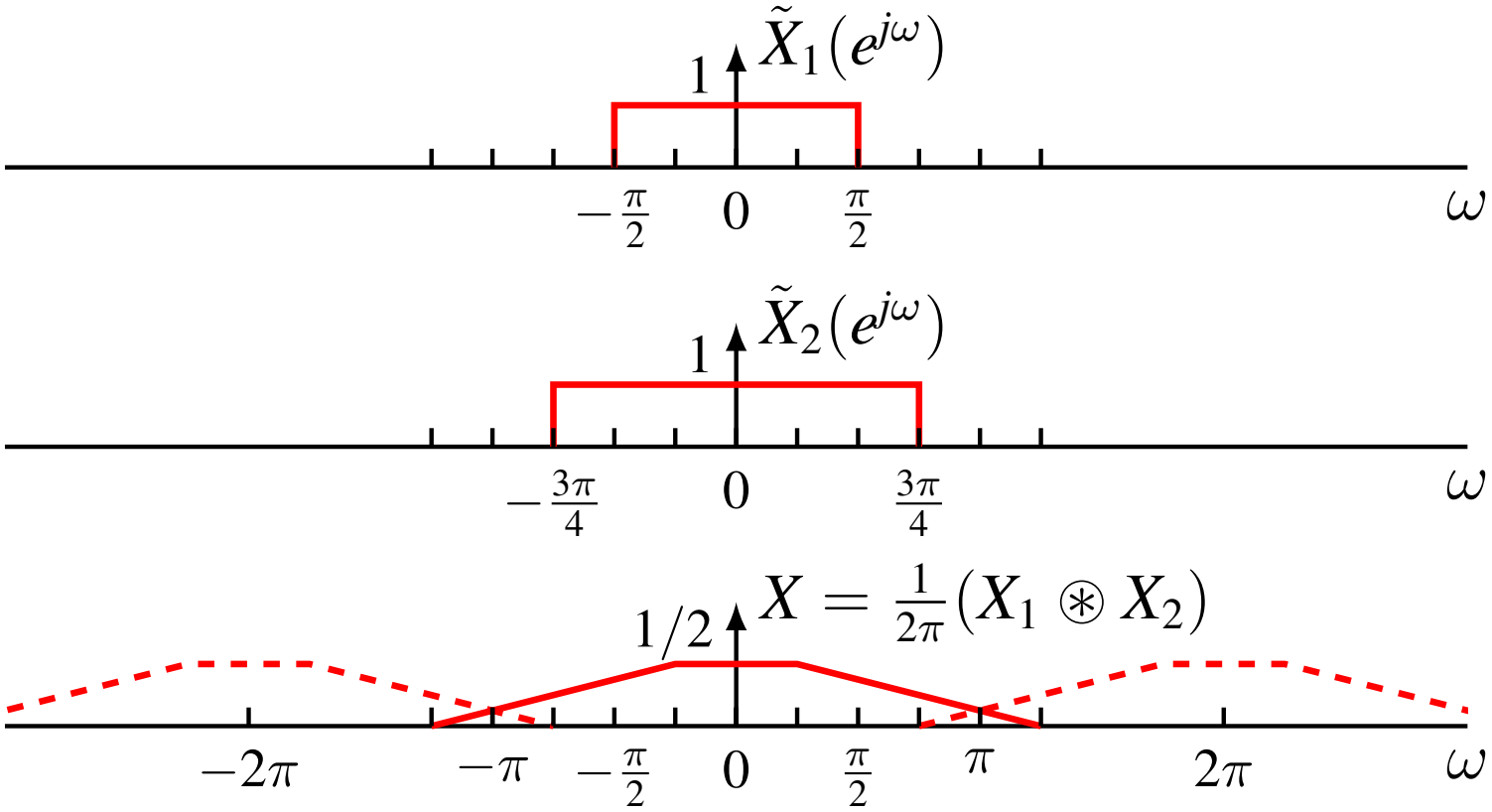

Example

$$x[n] = x_1[n]x_2[n], \text{ where } x_1[n] = \frac{\sin(3\pi n/4)}{\pi n}, \; x_2[n] = \frac{\sin(\pi n/2)}{\pi n}$$

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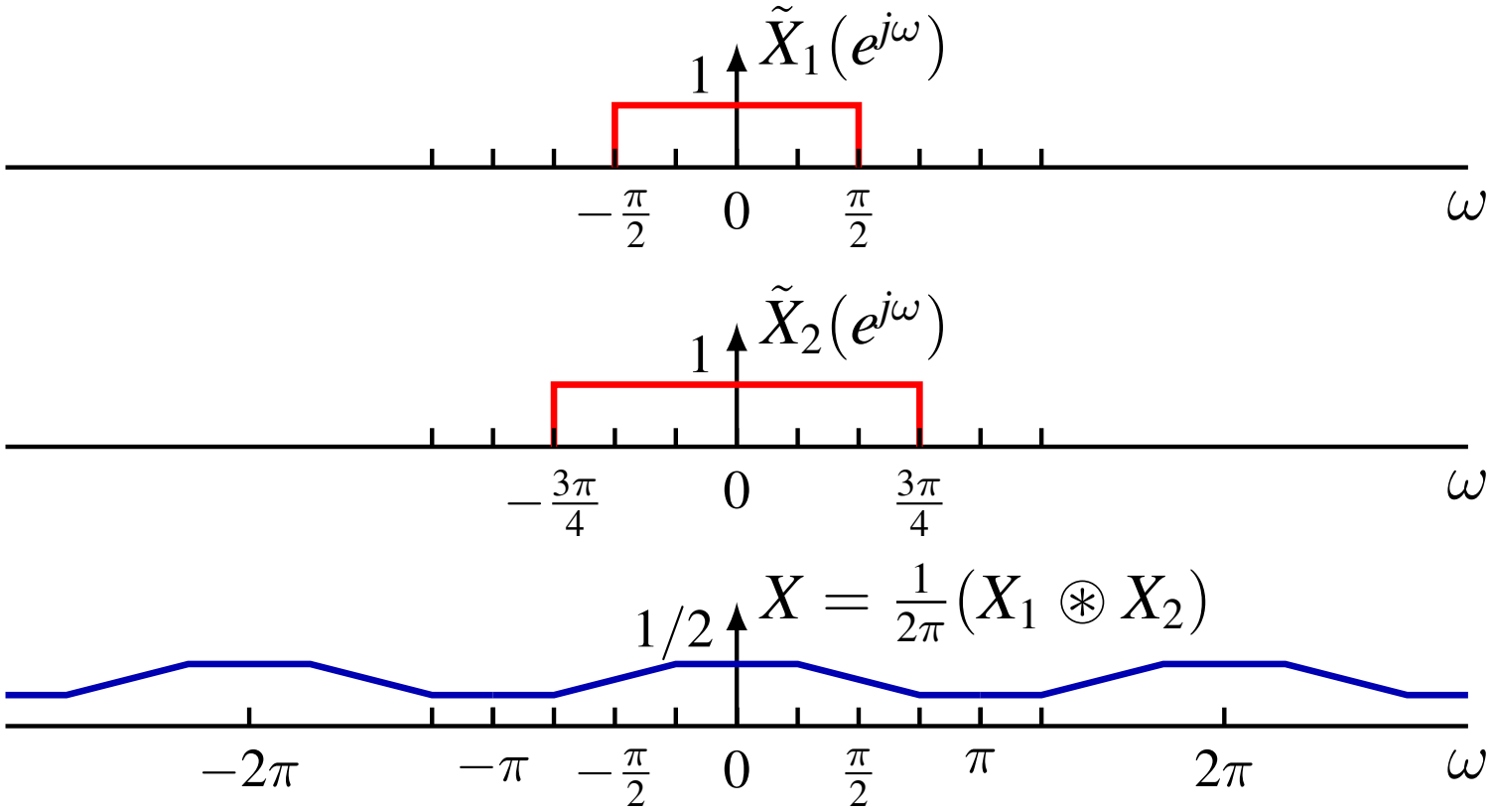
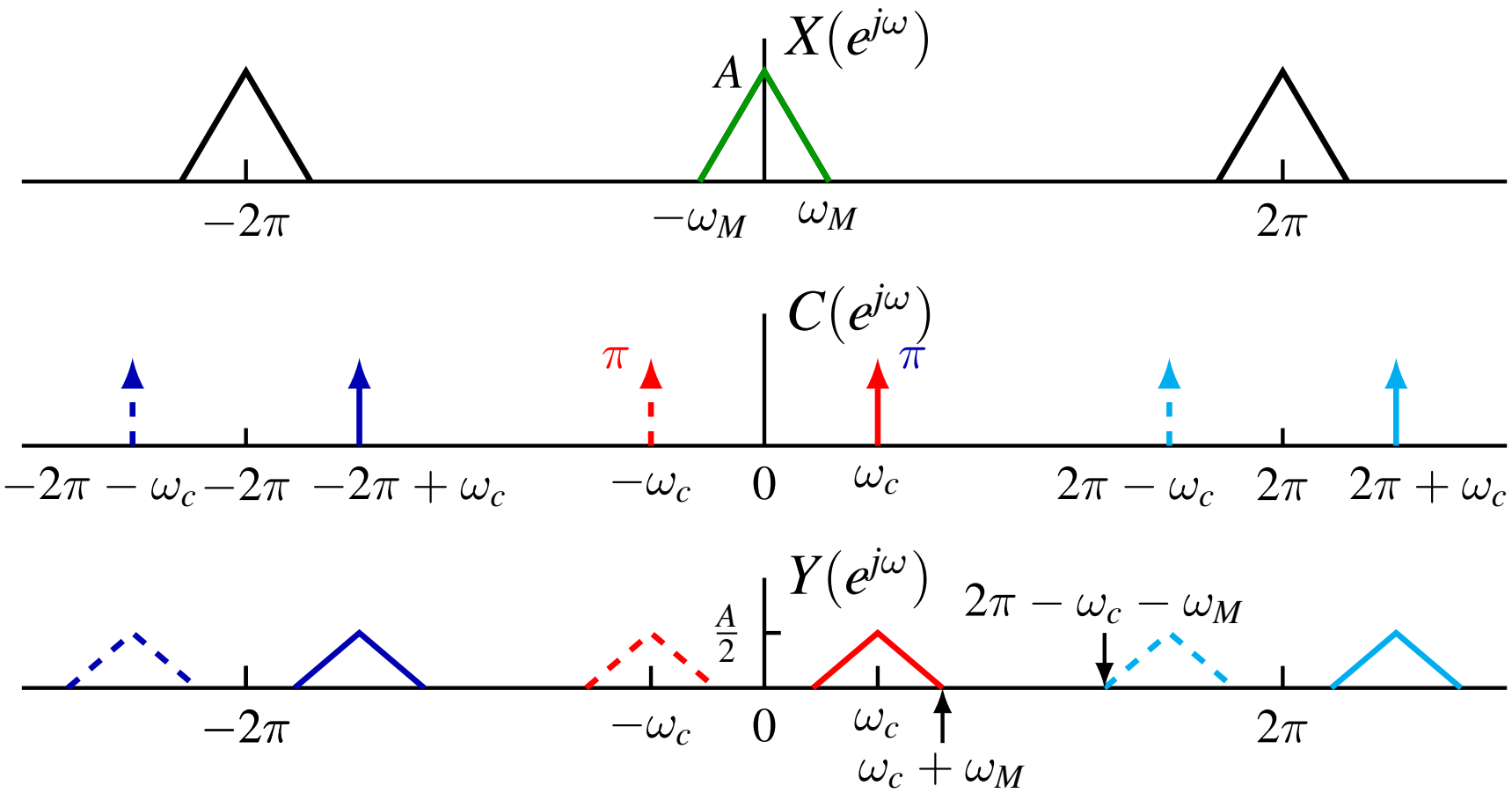


Example: DT Modulation

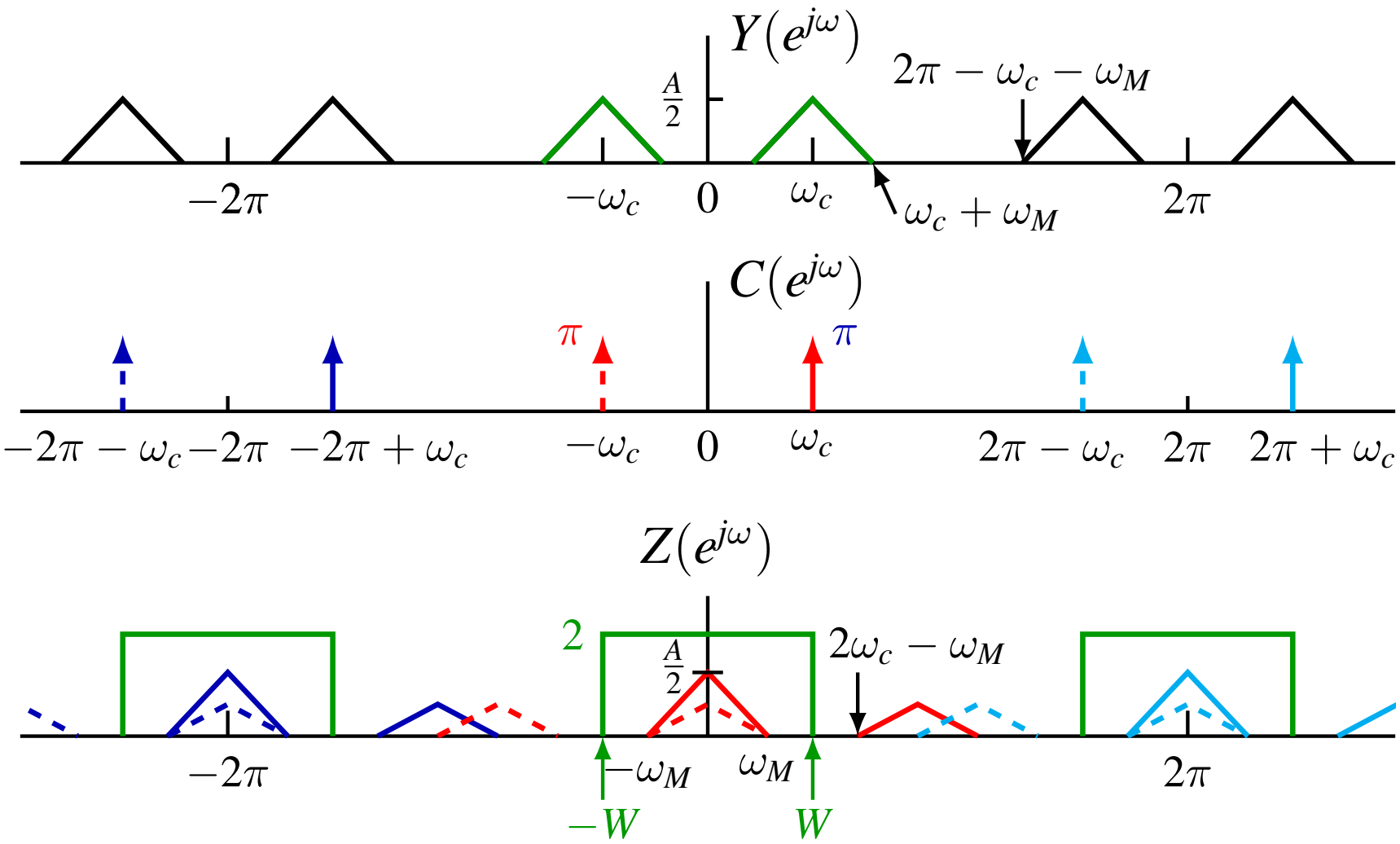
$$y[n] = x[n]c[n], \text{ where } c[n] = \cos(\omega_c n)$$



No overlap between replicas: $\begin{cases} \omega_c > \omega_M \\ \omega_c + \omega_M < \pi \end{cases} \implies \omega_M < \frac{\pi}{2}$

Example: DT Demodulation

$$z[n] = y[n]c[n], \text{ where } c[n] = \cos(\omega_c n)$$



Recover X by lowpass filtering Y with $W \in (\omega_M, 2\omega_c - \omega_M)$

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Linear Constant-coefficient Difference Equations

Frequency response of LTI system described by

$$\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$$

Method 1. Use eigenfunction property

$$x[n] = e^{j\omega n} \implies y[n] = H(e^{j\omega})e^{j\omega n}$$

Substitution into difference equation yields

$$\sum_{k=0}^{N} a_k H(e^{j\omega})e^{j\omega(n-k)} = \sum_{k=0}^{M} b_k e^{j\omega(n-k)}$$

$$H(e^{j\omega}) = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

Linear Constant-coefficient Difference Equations

Method 2. Take Fourier transform of both sides

$$\mathcal{F}\left\{\sum_{k=0}^{N} a_k y[n-k]\right\} = \mathcal{F}\left\{\sum_{k=0}^{M} b_k x[n-k]\right\}$$

By linearity and time-shifting property

$$\sum_{k=0}^{N} a_k e^{-j\omega k} Y(e^{j\omega}) = \sum_{k=0}^{M} b_k e^{-j\omega k} X(e^{j\omega})$$

By convolution property

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\sum_{k=0}^{M} b_k e^{-j\omega k}}{\sum_{k=0}^{N} a_k e^{-j\omega k}}$$

Frequency response is rational function of $e^{-j\omega}$

Example

Consider LTI system described by

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = 2x[n]$$

Frequency response

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}$$

By partial fraction expansion

$$H(e^{j\omega}) = \frac{2}{(1 - \frac{1}{2}e^{-j\omega})(1 - \frac{1}{4}e^{-j\omega})} = \frac{4}{1 - \frac{1}{2}e^{-j\omega}} - \frac{2}{1 - \frac{1}{4}e^{-j\omega}}$$

Take inverse Fourier transform to obtain impulse response

$$h[n] = 4\left(\frac{1}{2}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n]$$

Example

Find response to $x[n] = \left(\frac{1}{4}\right)^n u[n]$.

Fourier transform of response

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) = \frac{2}{(1-\frac{1}{2}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})} \cdot \frac{1}{1-\frac{1}{4}e^{-j\omega}}$$

By partial fraction expansion

$$Y(e^{j\omega}) = -\frac{4}{1-\frac{1}{4}e^{-j\omega}} - \frac{2}{(1-\frac{1}{4}e^{-j\omega})^2} + \frac{8}{1-\frac{1}{2}e^{-j\omega}}$$

Take inverse Fourier transform to obtain response

$$y[n] = \left\{-4\left(\frac{1}{4}\right)^n - 2(n+1)\left(\frac{1}{4}\right)^n + 8\left(\frac{1}{2}\right)^n\right\} u[n]$$

Partial Fraction Expansion

$$H(e^{j\omega}) = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}$$

Replace $e^{-j\omega}$ by z,

$$H(z^{-1}) = \frac{2}{1 - \frac{3}{4}z + \frac{1}{8}z^2} = \frac{2}{(1 - \frac{1}{2}z)(1 - \frac{1}{4}z)} = \frac{A}{1 - \frac{1}{2}z} + \frac{B}{1 - \frac{1}{4}z}$$

$$A = \left[(1 - \frac{1}{2}z)H(z^{-1})\right]\Big|_{z=2} = \frac{2}{1 - \frac{1}{4}z}\Big|_{z=2} = 4$$

$$B = \left[(1 - \frac{1}{4}z)H(z^{-1})\right]\Big|_{z=4} = \frac{2}{1 - \frac{1}{2}z}\Big|_{z=4} = -2$$

NB. Terms take form $\frac{A}{(s-r)^m}$ for CT, but $\frac{A}{(1-rz)^m}$ for DT

Partial Fraction Expansion

$$Y(e^{j\omega}) = \frac{2}{(1-\frac{1}{2}e^{-j\omega})(1-\frac{1}{4}e^{-j\omega})} \cdot \frac{1}{1-\frac{1}{4}e^{-j\omega}}$$

Replace $e^{-j\omega}$ by z,

$$Y(z^{-1}) = \frac{2}{(1-\frac{1}{2}z)(1-\frac{1}{4}z)^2} = \frac{A_{1,1}}{1-\frac{1}{2}z} + \frac{A_{2,1}}{1-\frac{1}{4}z} + \frac{A_{2,2}}{(1-\frac{1}{4}z)^2}$$

To find $A_{1,1}$

1. Multiply both side by $1-\frac{1}{2}z$,

$$(1-\frac{1}{2}z)Y(z^{-1}) = \frac{2}{(1-\frac{1}{4}z)^2} = A_{1,1} + \frac{A_{2,1}(1-\frac{1}{2}z)}{1-\frac{1}{4}z} + \frac{A_{2,2}(1-\frac{1}{2}z)}{(1-\frac{1}{4}z)^2}$$

2. Evaluate at $z=2$, i.e. root of $1-\frac{1}{2}z=0$

$$A_{1,1} = \left[(1-\frac{1}{2}z)Y(z^{-1})\right]\Big|_{z=2} = \frac{2}{(1-\frac{1}{4}z)^2}\Big|_{z=2} = 8$$

Partial Fraction Expansion

$$Y(z^{-1}) = \frac{2}{(1-\frac{1}{2}z)(1-\frac{1}{4}z)^2} = \frac{A_{1,1}}{1-\frac{1}{2}z} + \frac{A_{2,1}}{1-\frac{1}{4}z} + \frac{A_{2,2}}{(1-\frac{1}{4}z)^2}$$

To find $A_{2,2}$

1. Multiply both side by $(1-\frac{1}{4}z)^2$,

$$(1-\frac{1}{4}z)^2 Y(z^{-1}) = \frac{2}{1-\frac{1}{2}z} = \frac{A_{1,1}(1-\frac{1}{4}z)^2}{1-\frac{1}{2}z} + A_{2,1}(1-\frac{1}{4}z) + A_{2,2}$$

2. Evaluate at $z=4$, i.e. root of $1-\frac{1}{4}z=0$

$$A_{2,2} = \left[(1-\frac{1}{4}z)^2 Y(z^{-1})\right]\Bigg|_{z=4} = \frac{2}{1-\frac{1}{2}z}\Bigg|_{z=4} = -2$$

Partial Fraction Expansion

$$Y(z^{-1}) = \frac{2}{(1-\frac{1}{2}z)(1-\frac{1}{4}z)^2} = \frac{A_{1,1}}{1-\frac{1}{2}z} + \frac{A_{2,1}}{1-\frac{1}{4}z} + \frac{A_{2,2}}{(1-\frac{1}{4}z)^2}$$

To find $A_{2,1}$

1. Multiply both side by $(1-\frac{1}{4}z)^2$,

$$(1-\frac{1}{4}z)^2 Y(z^{-1}) = \frac{2}{1-\frac{1}{2}v} = \frac{A_{1,1}(1-\frac{1}{4}z)^2}{1-\frac{1}{2}z} + A_{2,1}(1-\frac{1}{4}z) + A_{2,2}$$

2. Differentiate both sides

$$\frac{d}{dz}(1-\frac{1}{4}z)^2 Y(z^{-1}) = \frac{1}{(1-\frac{1}{2}z)^2} = \frac{d}{dz}\frac{A_{1,1}(1-\frac{1}{4}z)^2}{1-\frac{1}{2}z} + \left(-\frac{\mathbf{1}}{\mathbf{4}}\right)A_{2,1}$$

3. Evaluate at $z=4$, i.e. root of $1-\frac{1}{4}z=0$

$$A_{2,1} = (-\mathbf{4})\left[\frac{d}{dz}(1-\frac{1}{4}z)^2 Y(z^{-1})\right]\Bigg|_{z=4} = \frac{-\mathbf{4}}{(1-\frac{1}{2}z)^2}\Bigg|_{z=4} = -4$$

Inverse Fourier Transform of $\frac{1}{(1-ae^{-j\omega})^n}$

Recall

$$a^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{1-ae^{-j\omega}}, \quad |a| < 1$$

i.e.

$$\frac{1}{1-az} = \sum_{n=0}^{\infty} a^n z^n, \quad z = e^{-j\omega}$$

Differentiate $m-1$ times w.r.t. z

$$\frac{(m-1)!a^{m-1}}{(1-az)^m} = \sum_{n=m-1}^{\infty} a^n \frac{n!}{(n-m+1)!} z^{n-(m-1)}$$

$$\frac{1}{(1-az)^m} = \sum_{n=m-1}^{\infty} a^{n-m+1} \binom{n}{m-1} z^{n-m+1} = \sum_{n=0}^{\infty} a^n \binom{n+m-1}{m-1} z^n$$

Thus

$$\binom{n+m-1}{n} a^n u[n] \xleftrightarrow{\mathcal{F}} \frac{1}{(1-ae^{-j\omega})^m}$$

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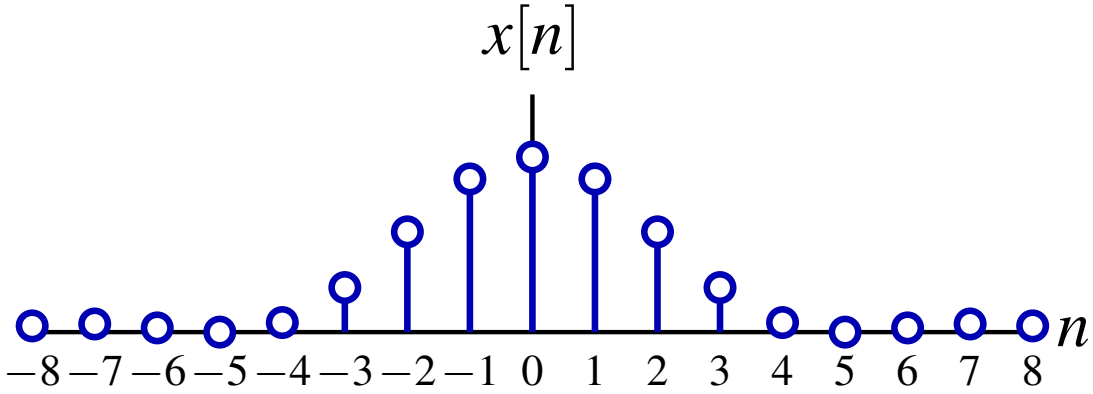
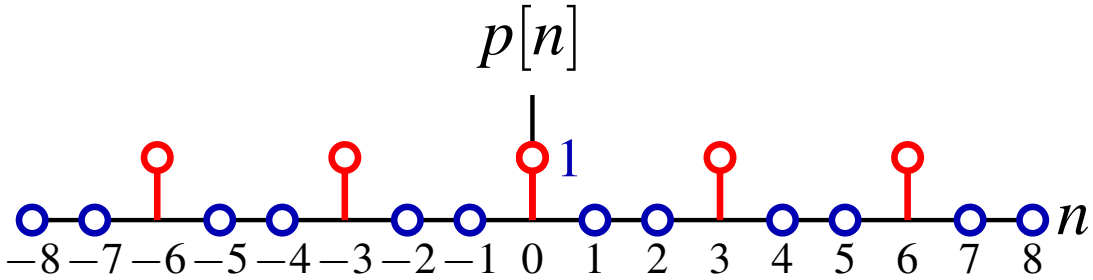
Impulse-train Sampling

DT signal $x[n]$



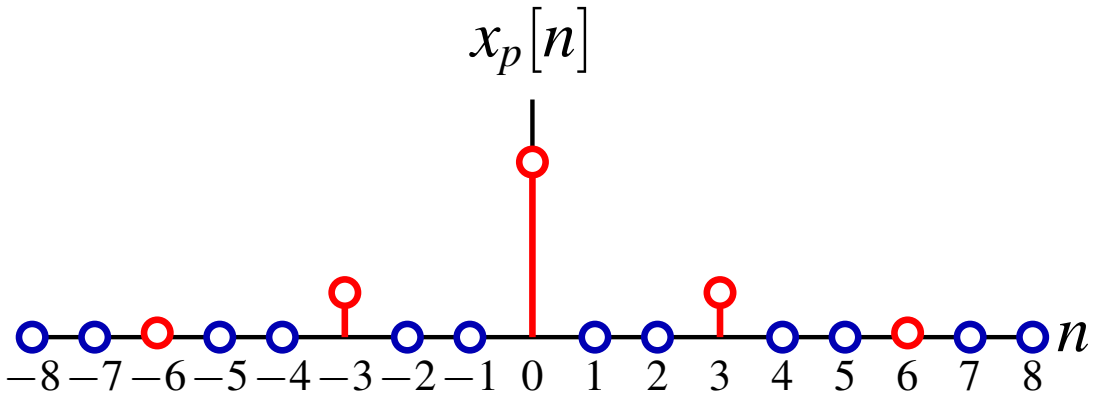
DT impulse train

$$p[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN]$$



Sampled signal

$$x_p[n] = x[n]p[n]$$
$$= \sum_{k=-\infty}^{\infty} x[kN]\delta[n - kN]$$



Impulse-train Sampling

Spectrum of impulse train

$$P(e^{j\omega}) = \omega_s \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s), \quad \omega_s = \frac{2\pi}{N}$$

Spectrum of sampled signal

$$X_p = \frac{1}{2\pi} X \circledast P = \frac{1}{2\pi} \tilde{X} * P = \frac{1}{2\pi} X * \tilde{P}$$

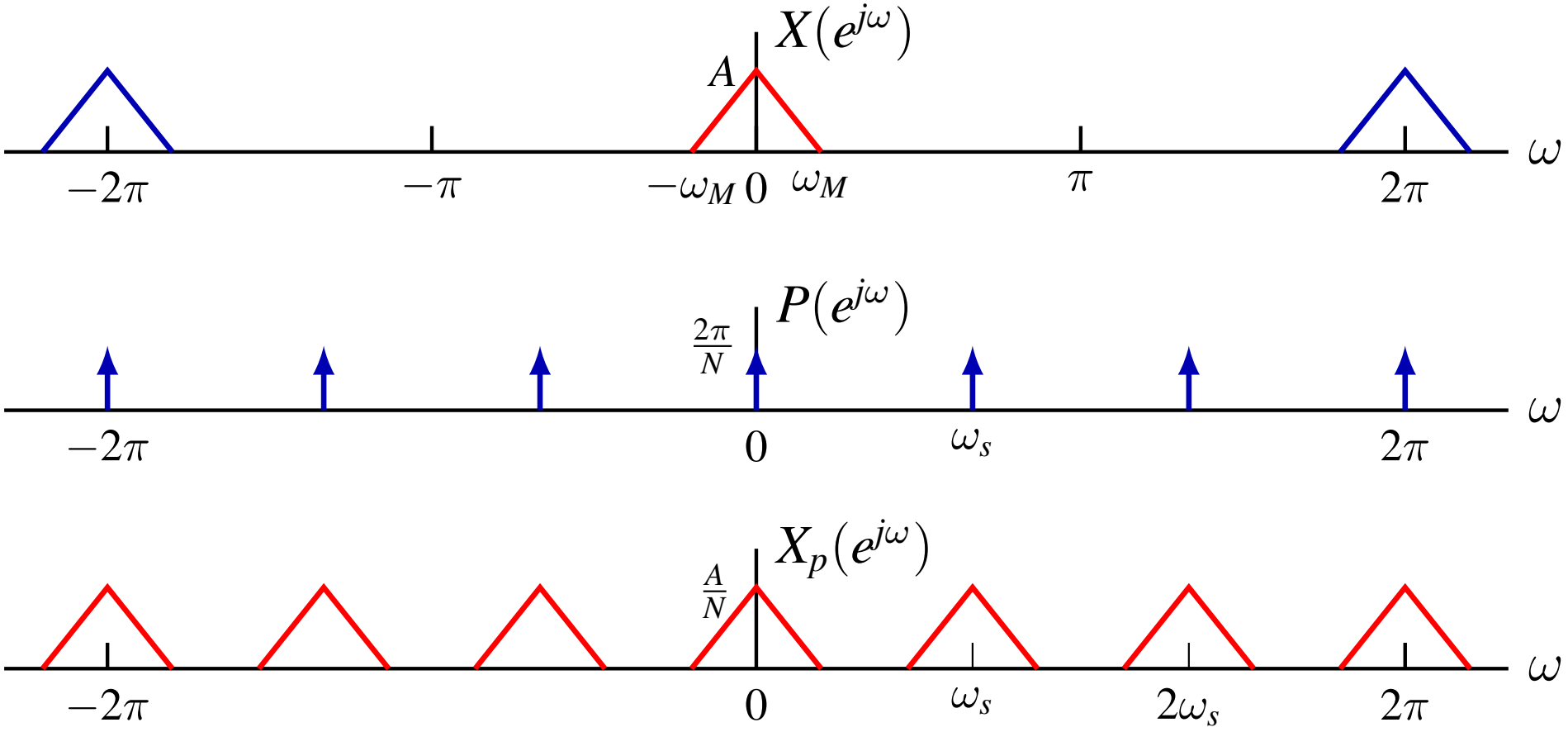
where $\tilde{X}$, $\tilde{P}$ are respective parts of X, P within one period of 2π

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=-\infty}^{\infty} \tilde{X}(e^{j(\omega - k\omega_s)}) = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$

Impulse-train Sampling

First view: Replicating $\tilde{X}$ at $k\omega_s$ for $k \in \mathbb{Z}$

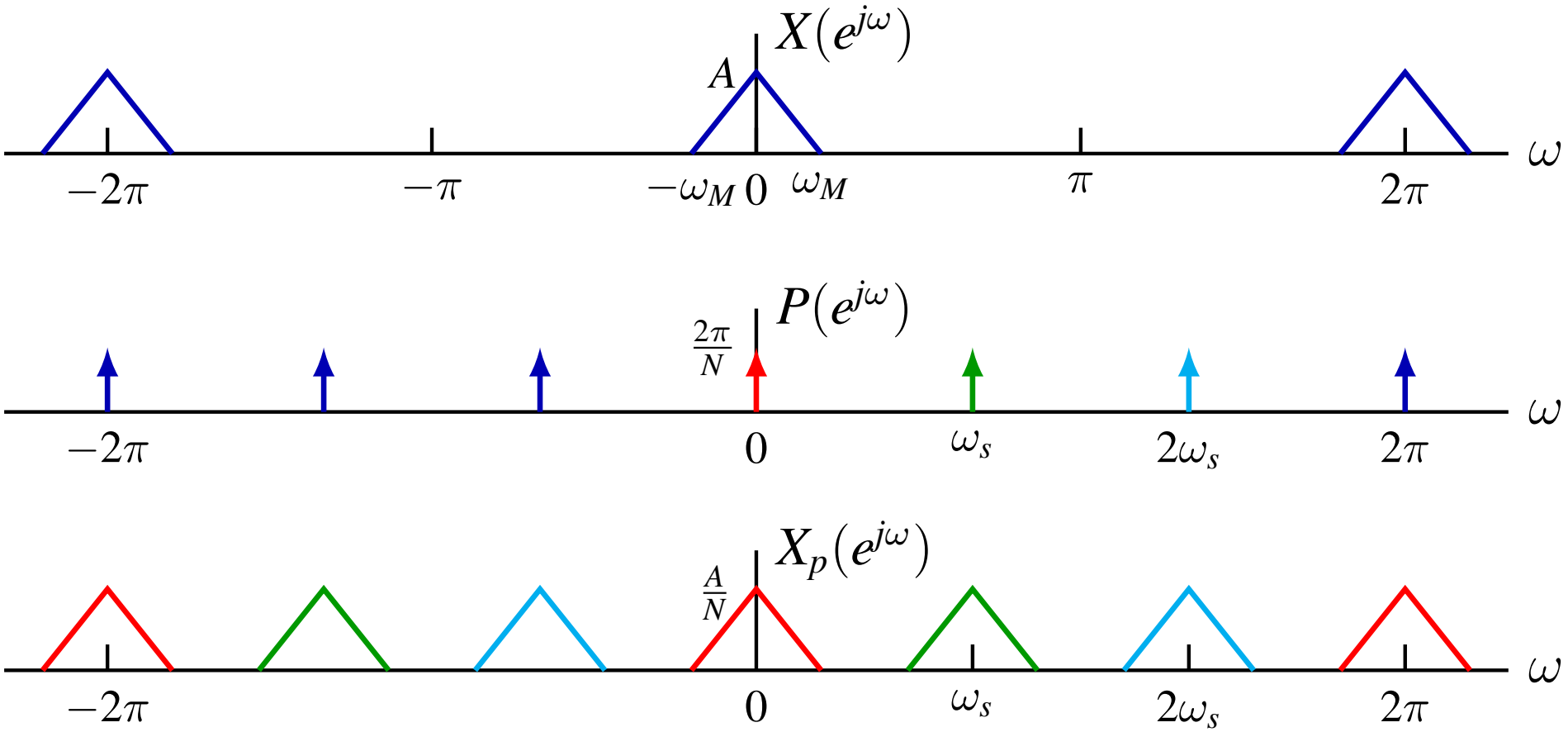
$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=-\infty}^{\infty} \tilde{X}(e^{j(\omega - k\omega_s)})$$



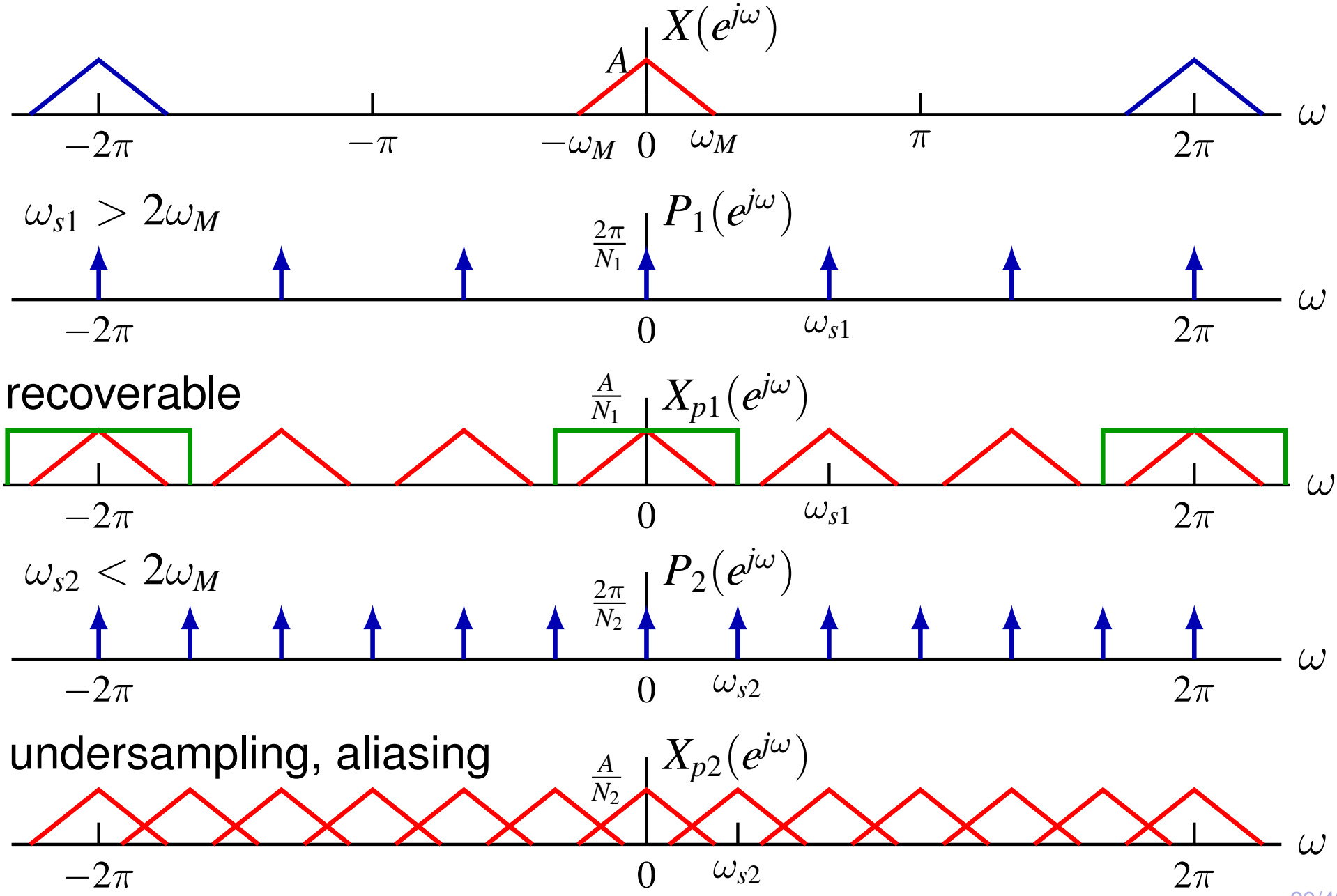
Impulse-train Sampling

Second view: Replicating X at $k\omega_s$ for $k = 0, 1, \ldots, N-1$

$$X_p(e^{j\omega}) = \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$



Impulse-train Sampling



DT Sampling Theorem

Band-limited DT signal $x[n]$ whose spectrum $X(e^{j\omega}) = 0$ for $\omega_M < |\omega| \leq \pi$ is uniquely determined by its samples $x[kN], k \in \mathbb{Z}$ if

$$\omega_s \triangleq \frac{2\pi}{N} > 2\omega_M$$

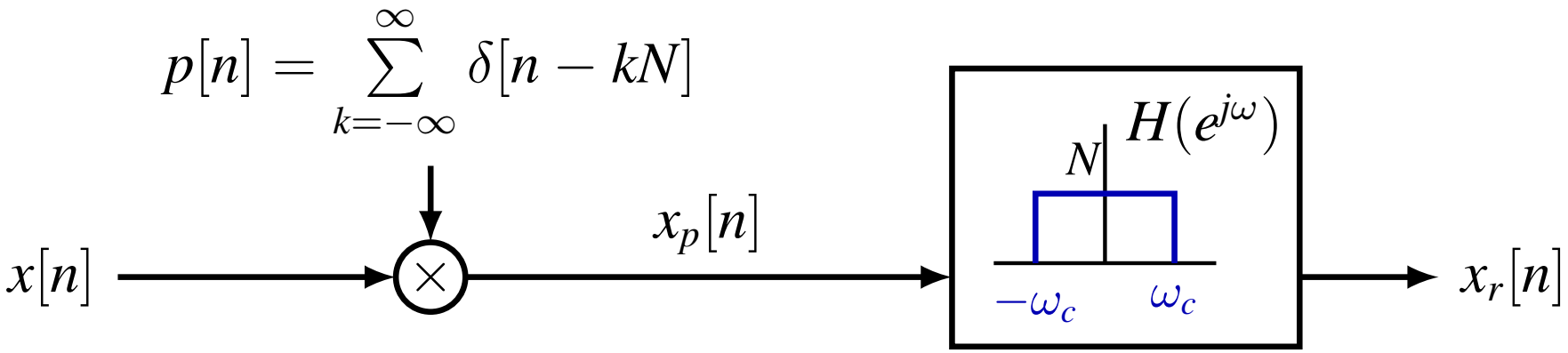
Given $\{x[kN] : k \in \mathbb{Z}\}$, $x[n]$ can be reconstructed as follows

1. construct $x_p[n] = \sum_{k=-\infty}^{\infty} x[kN]\delta[n - kN]$
2. send x_p through lowpass filter with gain N and cutoff frequency $\omega_c \in (\omega_M, \omega_s - \omega_M)$, i.e.

$$H(e^{j\omega}) = N[u(\omega + \omega_c) - u(\omega - \omega_c)]$$

3. output $x_r[n]$ with $X_r(e^{j\omega}) = X_p(e^{j\omega})H(e^{j\omega})$ is same as $x[n]$

Reconstruction in Time Domain



Impulse response of lowpass filter

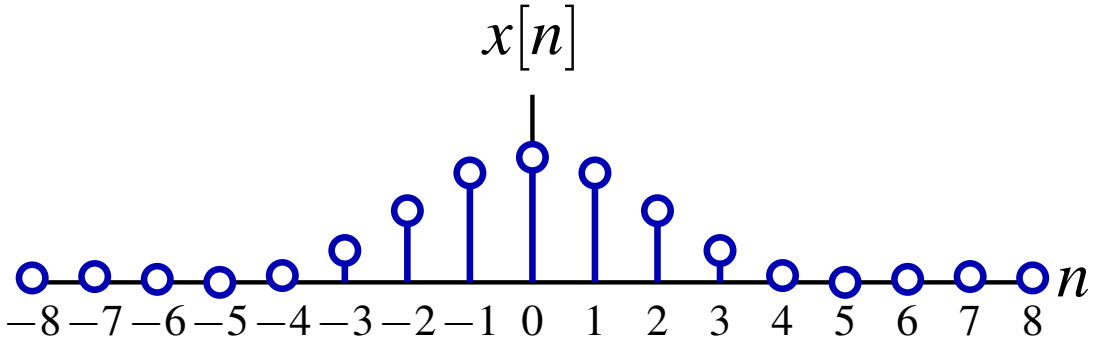
$$h[n] = \frac{N \sin(\omega_c n)}{\pi n}$$

x recovered by band-limited interpolation using sinc function

$$x[n] = x_r[n] = (x_p * h)[n] = \sum_{k=-\infty}^{\infty} x[kN] \frac{N \sin(\omega_c (n-kN))}{\pi (n-kN)}$$

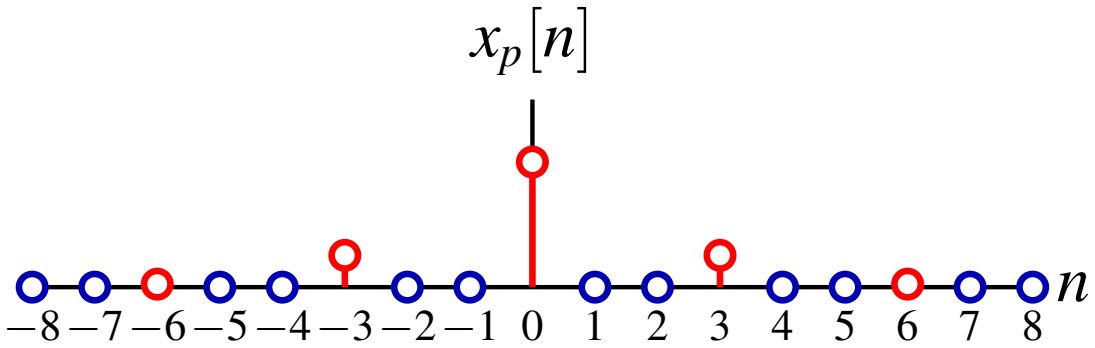
Decimation

DT signal $x[n]$



Sampled signal

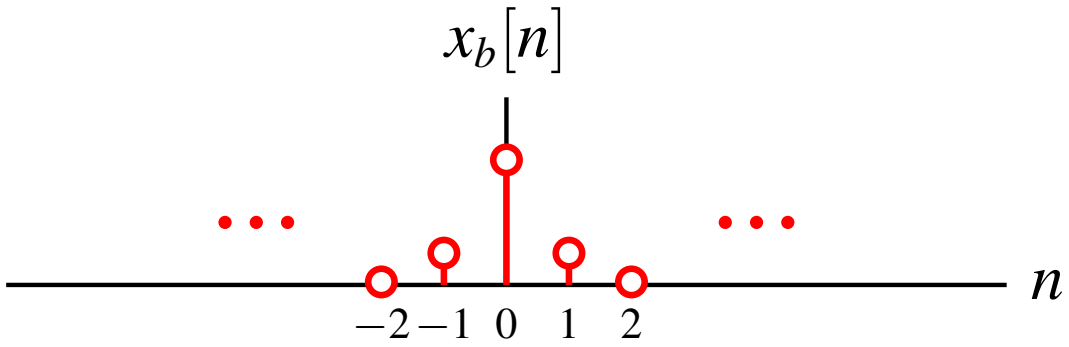
$$x_p[n] = \sum_{k=-\infty}^{\infty} x[kN]\delta[n-kN]$$



Decimated signal

$$x_b[n] = x[nN]$$

Reduces samples



Decimation

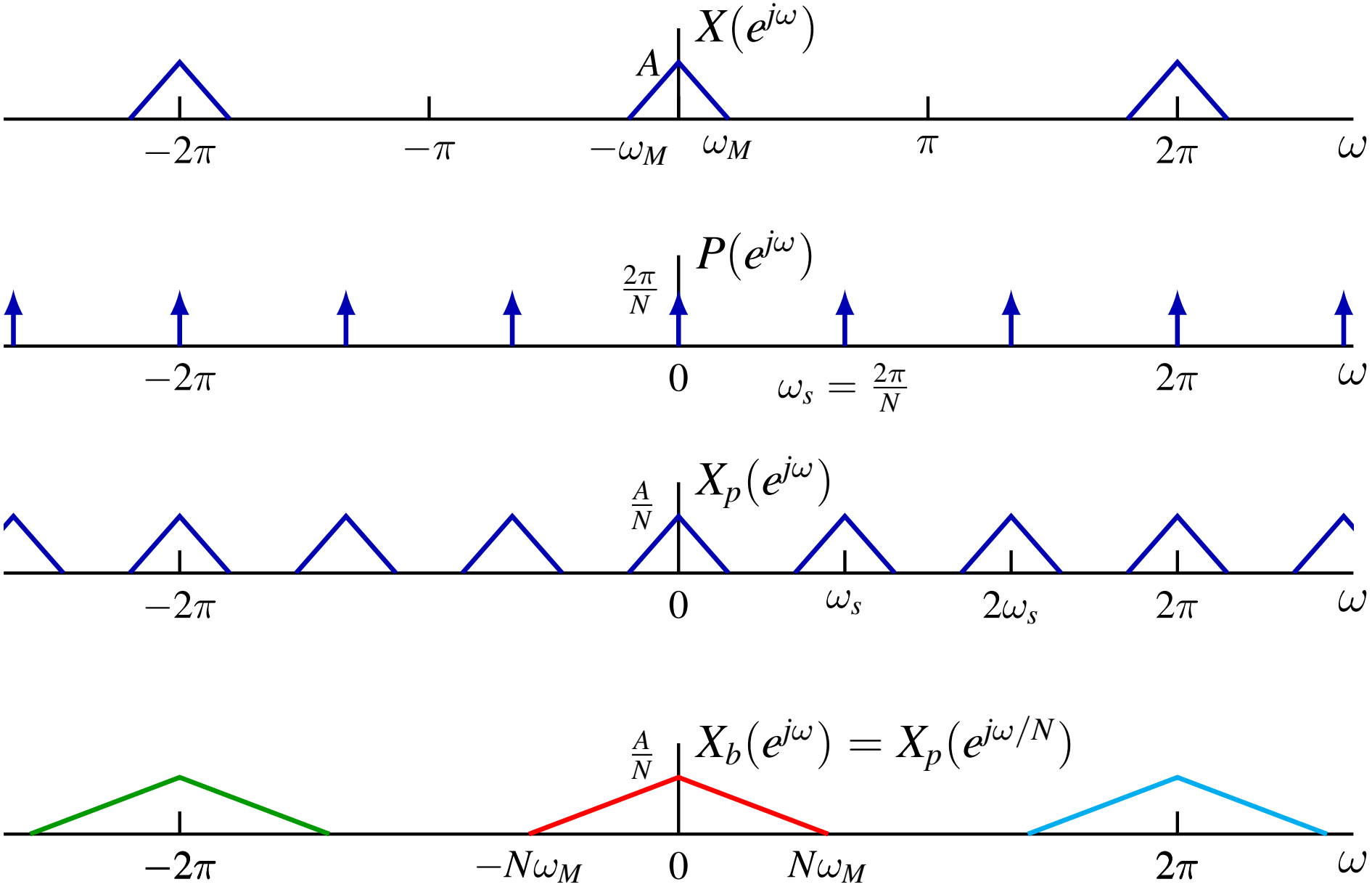
$$
\begin{aligned}
X_b(e^{j\omega}) &= \sum_{n\in\mathbb{Z}} x_b[n]e^{-j\omega n} \\
&= \sum_{n\in\mathbb{Z}} x[nN]e^{-j\omega n} \\
&= \sum_{k\in N\mathbb{Z}} x[k]e^{-j\omega k/N} \quad (k = nN) \\
&= \sum_{k\in\mathbb{Z}} x_p[k]e^{-j\frac{\omega}{N}k} \quad (x_p[n] = 0 \text{ for } n \in \mathbb{Z}\setminus N\mathbb{Z}) \\
&= X_p(e^{j\omega/N})
\end{aligned}
$$

Recall $X_p(e^{j\omega})$ has period $2\pi/N$, so $X_b(e^{j\omega)}$ has period 2π.

If no aliasing in sampling

- X_b is stretched version of X **in each period**
- $X_b(e^{j\omega}) = X_p(e^{j\omega/N})$, but $X_b(e^{j\omega}) \neq X(e^{j\omega/N})$

Decimation


$X(e^{j\omega})$
A
-2π
$-\pi$
$-\omega_M$
ω_M
π
2π
ω
$P(e^{j\omega})$
$\frac{2\pi}{N}$
-2π
0
$\omega_s = \frac{2\pi}{N}$
2π
ω
$X_p(e^{j\omega})$
$\frac{A}{N}$
-2π
0
ω_s
$2\omega_s$
2π
ω
$X_b(e^{j\omega}) = X_p(e^{j\omega/N})$
$\frac{A}{N}$
-2π
$-N\omega_M$
0
$N\omega_M$
2π
ω

Contents

1. Convolution Property of DTFT

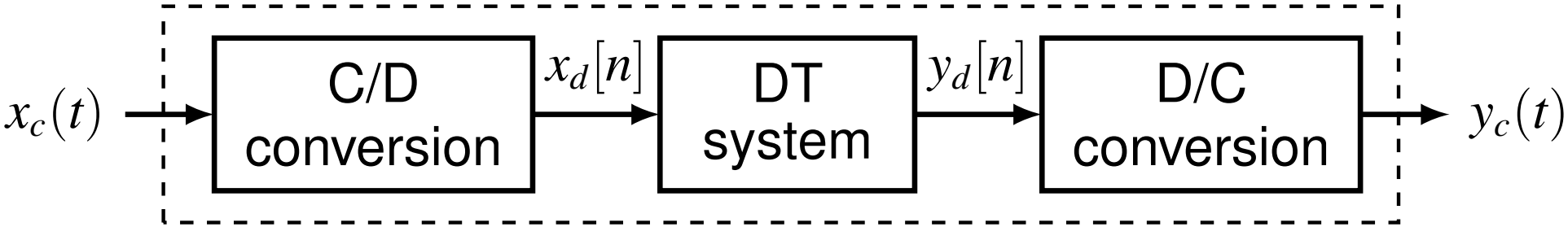
2. Multiplication Property of DTFT

3. Systems Described by Linear Constant-coefficient Difference Equations

4. Sampling of DT Signals

5. DT Processing of CT Signals

DT Processing of CT Signals



Benefits of DT processing

- versatility of digital electronic devices
- resilience against noise

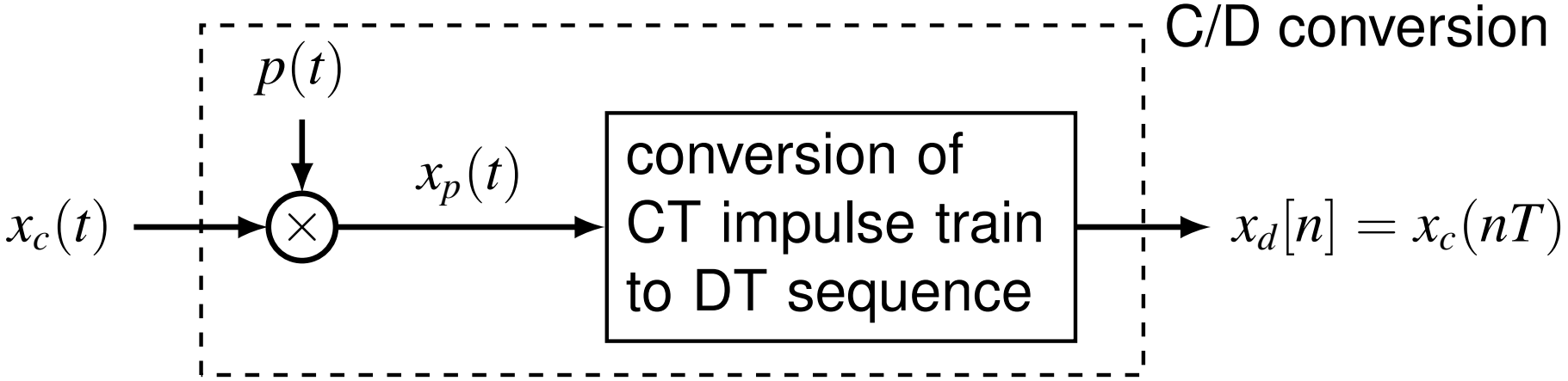
In practice

- C/D implemented by analog-to-digital converter
- D/C implemented by digital-to-analog converter

Idealization

- no quantization
- C/D by impulse-train sampling
- D/C by ideal reconstruction

C/D Conversion



Assumption

- $x_c(t)$ bandlimited, e.g. lowpass filtered
- sampling above Nyquist rate, no aliasing

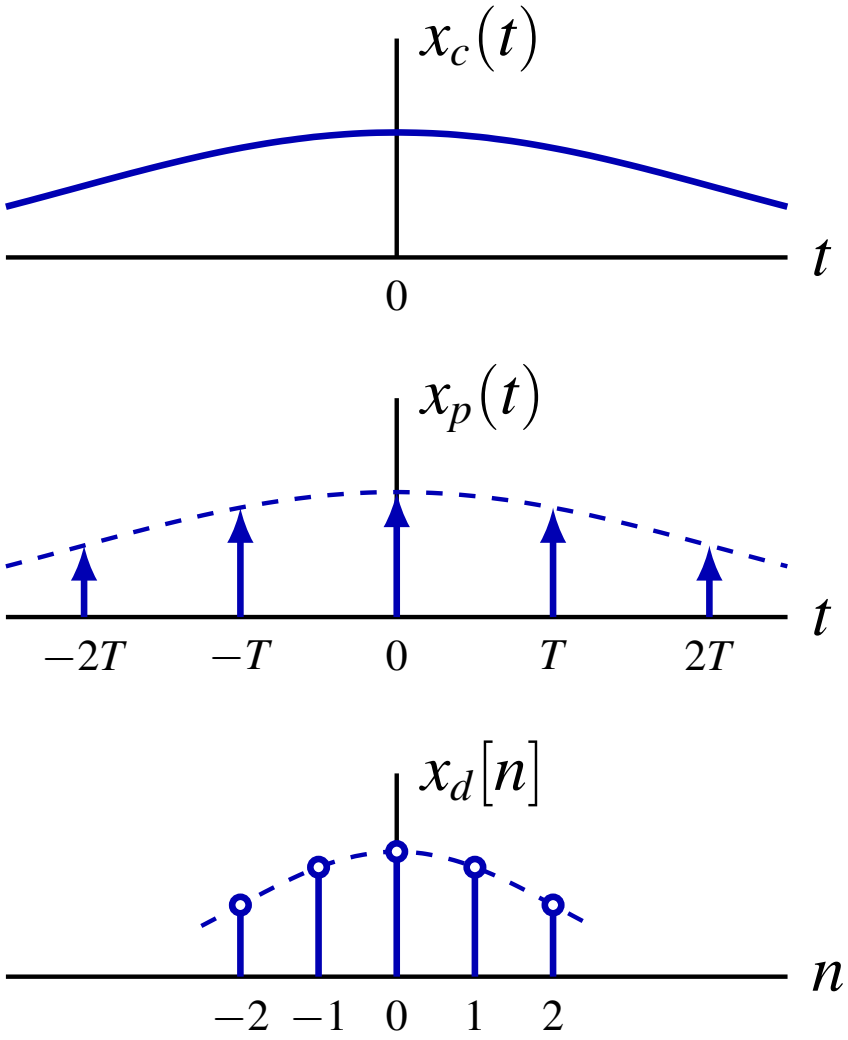
Relation between CT and DT spectra

$$X_p(j\omega) = \text{CTFT}\left\{\sum_{n\in\mathbb{Z}} x_c(nT)\delta(t-nT)\right\}$$

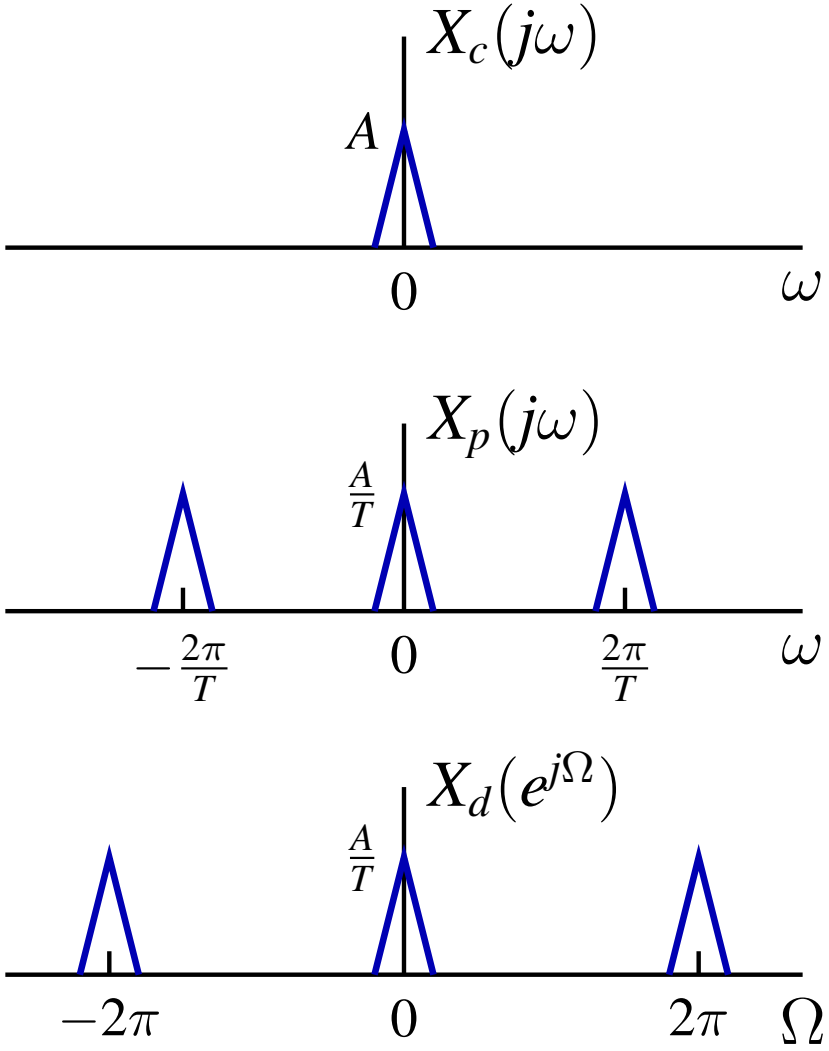
$$= \sum_{n\in\mathbb{Z}} x_c(nT)e^{-j\omega Tn} = \sum_{n\in\mathbb{Z}} x_d[n]e^{-j\omega Tn} = X_d(e^{j\omega T})$$

C/D Conversion

Time domain

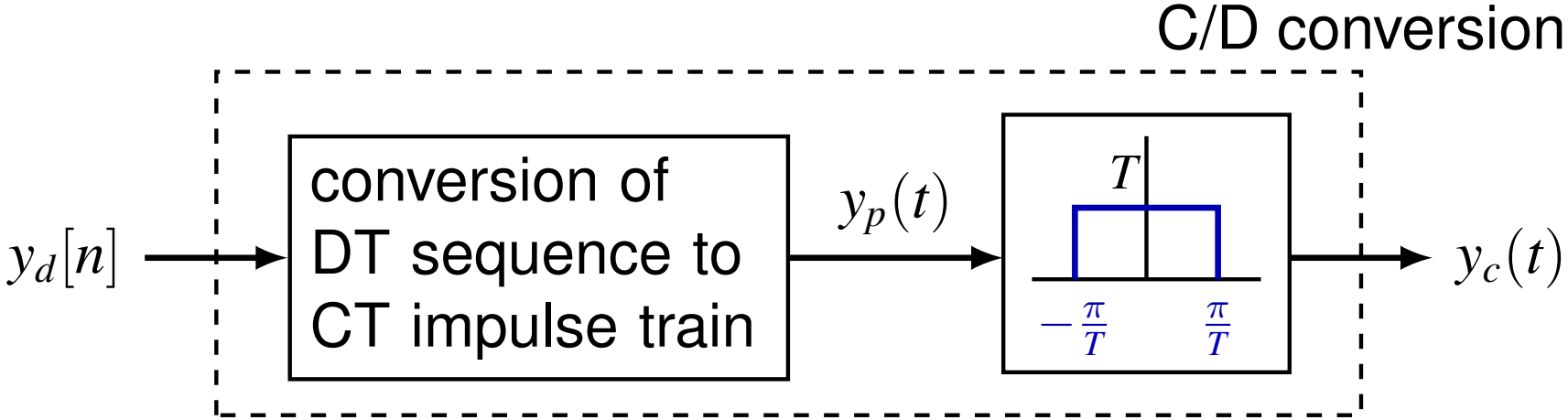


Frequency domain



From x_p to x_d: normalization in both time and frequency

D/C Conversion



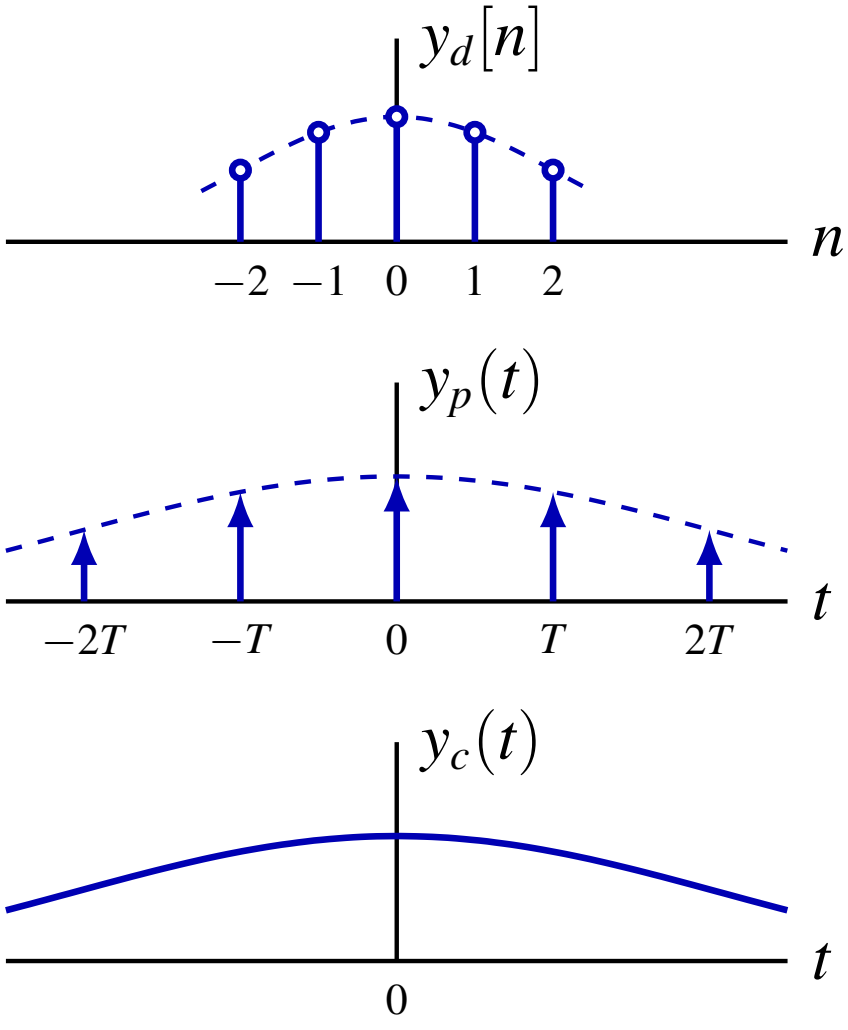
Relation between CT and DT spectra

$$Y_p(j\omega) = Y_d(e^{j\omega T})$$

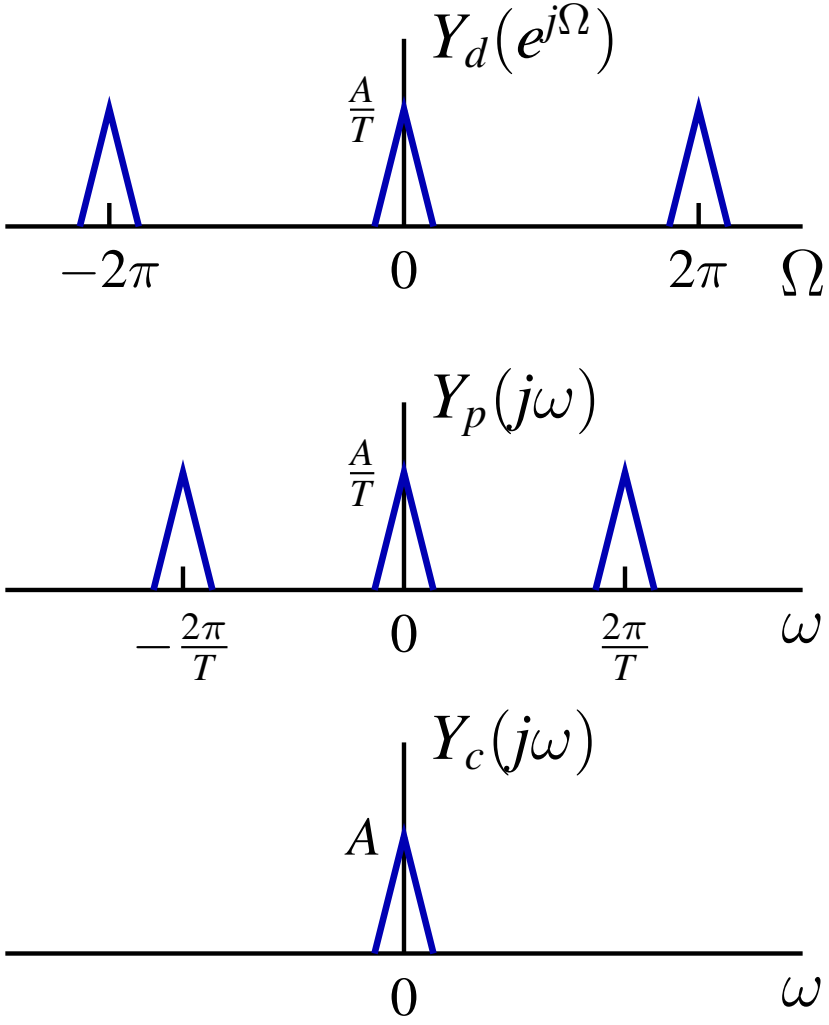
$$Y_c(j\omega) = \begin{cases} Y_p(j\omega) & |\omega| < \dfrac{\omega_s}{2} = \dfrac{\omega}{T} \\ 0, & \text{otherwise} \end{cases}$$

D/C Conversion

Time domain

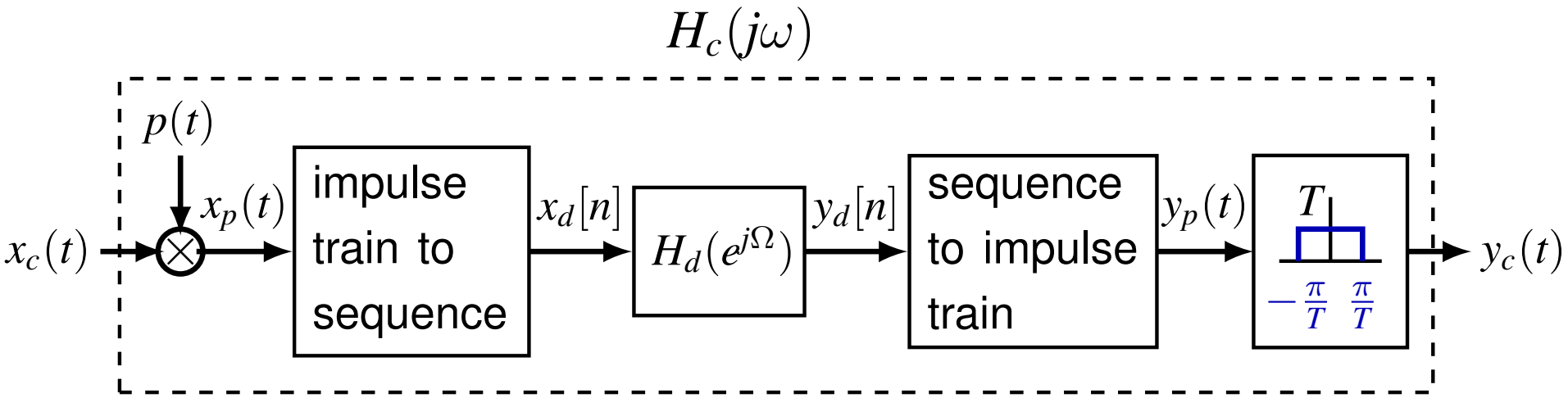


Frequency domain



Reverse process of C/D conversion

Frequency Response

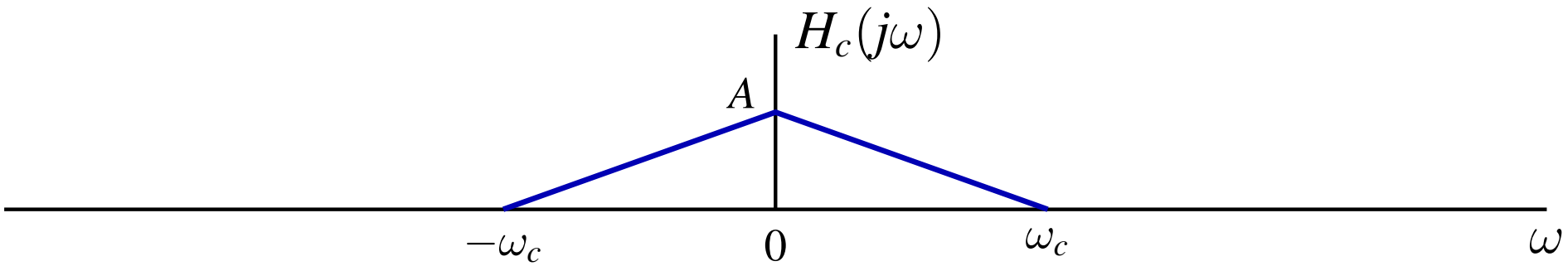


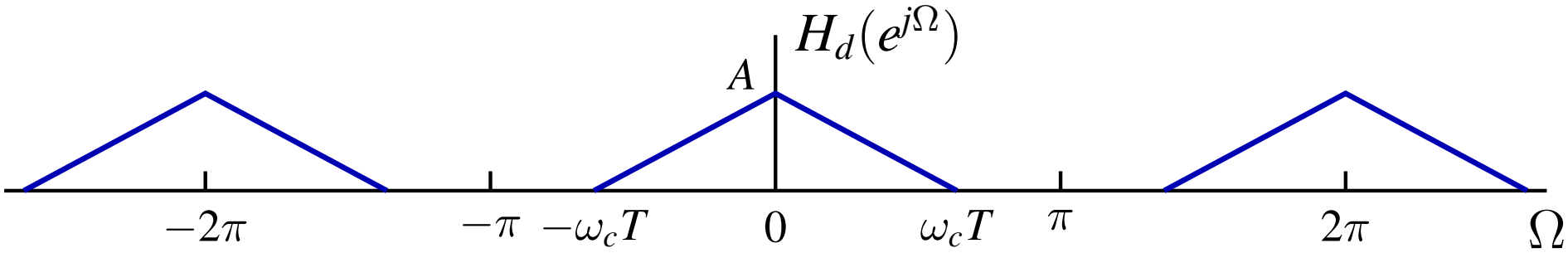
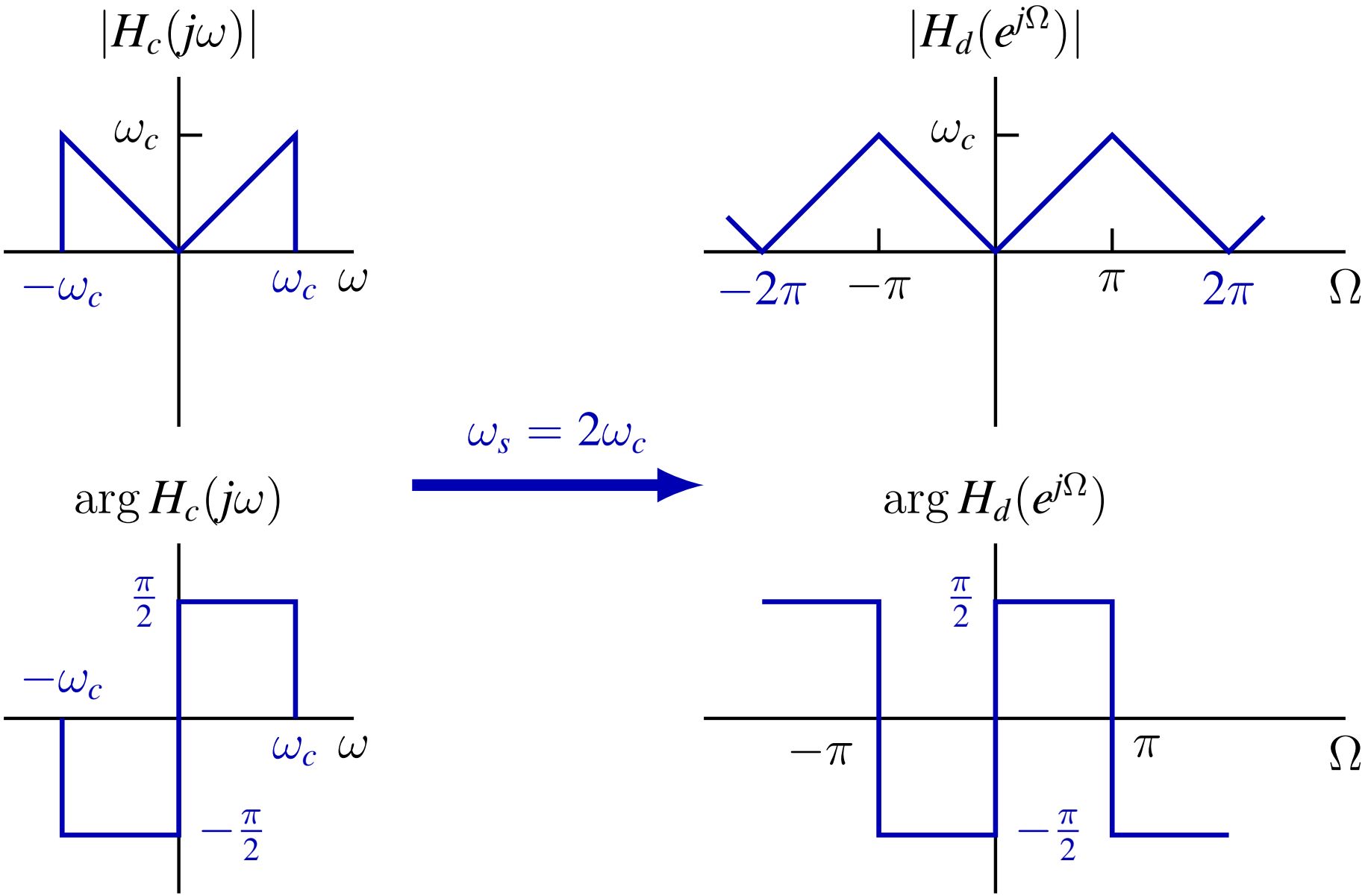
1. $Y_d(e^{j\Omega}) = H_d(e^{j\Omega})X_d(e^{j\Omega})$
2. $Y_p(j\omega) = Y_d(e^{j\omega T}) = H_d(e^{j\omega T})X_d(e^{j\omega T}) = H_d(e^{j\omega T})X_p(j\omega)$
3. $Y_c(j\omega) = Y_p(j\omega)H_{\text{lp}}(j\omega) = H_d(e^{j\omega T})X_p(j\omega)H_{\text{lp}}(j\omega) = H_d(e^{j\omega T})X_c(j\omega)$
4. $H_c(j\omega) = \begin{cases} H_d(e^{j\omega T}) & |\omega| < \dfrac{\omega_s}{2} = \dfrac{\omega}{T} \\ 0, & \text{otherwise} \end{cases}$

Frequency Response

$$H_c(j\omega) = \begin{cases} H_d(e^{j\omega T}) & |\omega| < \dfrac{\omega_s}{2} = \dfrac{\omega}{T} \\ 0, & \text{otherwise} \end{cases}$$

$$\implies H_d(e^{j\Omega}) = H_c\left(j\frac{\Omega}{T}\right), |\Omega| < \pi$$





Example: Digital Differentiator